

Project list of CSE 3102

CO2-PO3-WK5

1. Moving Car Animation System

Draw a road using lines and animate a car moving from one side of the screen to the other. Use circles for wheels and rotate them continuously. Add trees, traffic lights, for a realistic road environment.

ID- 231311004, 241311041, 232311102, 231311012, 241311175, 241311181, 241311119

2. Analog Clock Simulation

Create a circular clock face with numbers and animate the hour, minute, and second hands according to real-time or system time. Use rotation transformation for hand movement.

ID- 231311011, 231311039, 241311042, 232311193, 231311029

3. Solar System Animation

Draw the sun and planets using circles. Animate planets revolving around the sun in circular paths with different orbital speeds. Add rotation effects to planets for realism.

ID- 231311100, 241311029, 241311043, 232311224, 231311108

4. Bouncing Ball Animation

Simulate a ball bouncing inside a window or on the ground. Use gravity-like motion and reverse direction when the ball hits boundaries.

ID- 232311011, 241311030, 241311045, 241311081, 231311120

5. Flying Bird Animation

Draw a bird using lines and curves. Animate wing flapping using rotation transformations while the bird moves across the sky with clouds and mountains.

ID- 232311014, 241311032, 241311047, 241311082, 232311146

6. Windmill Rotation Project

Design a windmill using line segments and polygons. Continuously rotate the blades around the center point using 2D rotation matrices.

ID- 232311029, 241311033, 241311049, 241311084, 232311206

7. Traffic Signal Simulation

Create a road intersection with traffic lights that change colors automatically. Vehicles stop at red lights and move during green lights.

ID- 232311055, 241311035, 241311050, 241311086, 232311332

8. Train Movement Animation

Draw railway tracks and animate a train moving smoothly across them. Rotate train wheels and optionally add smoke effects from the engine.

ID- 232311059, 241311036, 241311051, 241311087, 241311121

9. Helicopter Flying Animation

Create a helicopter using polygons and circles. Rotate the top and tail rotors continuously while the helicopter moves through the sky.

ID- 241311001, 241311037, 241311052, 241311089, 241311122

10. Rocket Launch Simulation

Draw a rocket standing on a launch pad. Animate upward movement with fire and smoke effects during launch.

ID- 241311002, 241311038, 241311053, 241311090, 241311125

11. Rainfall Animation System

Simulate rainfall using multiple animated line segments falling from top to bottom. Add clouds and occasional lightning effects.

ID- 241311003, 241311039, 241311056, 241311091, 241311127

12. Snowfall Animation Project

Create snowflakes using small circles or points that slowly fall downward with random movement patterns.

ID- 241311004, 241311163, 241311057, 241311092, 241311128

13. Water Wave Animation

Draw moving sinusoidal wave patterns to simulate sea or river water motion. Add boats or floating objects for extra animation.

ID- 241311005, 241311058, 241311093, 241311129, 241311157

14. Fish Aquarium Simulation

Create an aquarium scene with swimming fish, bubbles rising upward, and animated plants swaying underwater.

ID- 241311006, 241311059, 241311094, 241311130, 241311158

15. Snake Movement Animation

Simulate a snake moving in curved motion across the screen using connected circles or line segments for the body.

ID- 241311007, 241311060, 241311095, 241311131, 241311159

16. Human Walking Animation

Design a stickman using lines and circles. Animate walking by rotating arms and legs alternately.

ID- 241311009, 241311062, 241311096, 241311132, 241311160

17. Pendulum Motion Simulation

Draw a pendulum attached to a fixed point and animate its oscillation using angular motion.

ID- 241311011, 241311063, 241311098, 241311133, 241311192

18. Fan Rotation Animation

Create an electric fan with rotating blades. Add speed control functionality using keyboard input.

ID- 241311012, 241311065, 241311100, 241311134, 241311193

19. Bicycle Riding Animation

Draw a bicycle with rotating wheels and animate it moving across roads while pedals rotate simultaneously.

ID- 241311013, 241311069, 241311102, 241311136, 241311194

20. Airplane Flying Simulation

Create an airplane moving through the sky with moving clouds and runway animations for takeoff and landing.

ID- 241311015, 241311070, 241311103, 241311141

21. Car Racing Mini Project

Develop a simple racing game where the player controls a car to avoid obstacles and score points.

ID- 241311016, 241311071, 241311106, 241311142

22. Digital Drawing Board

Implement a drawing application where users can draw lines, circles, rectangles, and freehand sketches using the mouse.

ID- 241311017, 241311074, 241311107, 241311143

23. Radar Scanning Animation

Draw a circular radar screen with a rotating scanning line that detects and highlights moving objects.

ID- 241311018, 241311075, 241311108, 241311144

24. Shooting Star Animation

Animate stars moving rapidly across a night sky leaving glowing trails behind them.

ID- 241311019, 241311162, 241311109, 241311145

25. Fireworks Animation System

Simulate fireworks exploding into colorful particles spreading outward in circular patterns.

ID- 241311020, 241311170, 241311110, 241311147

26. Rotating Earth and Moon Simulation

Animate Earth rotating on its axis while the moon revolves around Earth in orbit.

ID- 241311023, 241311176, 241311111, 241311148

27. Elevator Movement Simulation

Create a building with multiple floors and animate an elevator moving between them with opening doors.

ID- 241311024, 232311149, 241311112, 241311149

28. ATM Interface Using Graphics

Design a graphical ATM interface with buttons, screen menus, PIN entry, and transaction simulation.

ID- 241311025, 232311181, 241311114, 241311153

29. Simple 2D Game Graphics Project

Build a game like Pong or Brick Breaker using collision detection, animation, and graphics primitives.

ID- 241311026, 232311188, 241311116, 241311154

30. Computer Graphics Paint Application

Create a paint program supporting drawing tools, color selection, erasing, shape drawing, and canvas clearing.

ID- 241311027, 232311190, 241311117, 241311156

Project Report Contents

1. Cover Page
2. Introduction
3. Problem Statement
 - Describe:
 - what problem exists
 - what your project addresses
4. Objectives
5. Algorithms Used
6. Implementation
 - Explain:
 - programming language used
 - graphics libraries/tools
7. Results/Outputs
8. Conclusion